UNIVERSITY OF AUCKLAND
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SCIENCE

COMPSCI 714
Lectutorial 1: Data basics

Data loading, exploring, cleaning and pre-processing

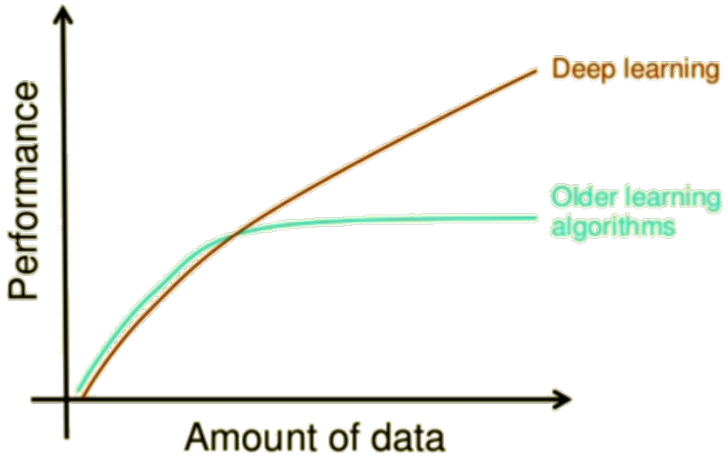
Instructor: Thomas Lacombe

Learning outcomes

- **Explain** the role of data in the development and performance of modern AI systems.
- **Identify** different types and sources of data.
- **Explain** the processes and considerations involved when acquiring and using data to train a model.
- **Recommend and carry out** exploratory data analysis to guide data cleaning and preprocessing choices.
- **Recommend and apply** basic data cleaning and preprocessing to prepare data for model training.

Data is fuel for AI and Machine Learning systems

- Training models with a lot of parameters (i.e., degrees of freedom) requires a lot of data.
- Common « 10 times » rule for deep neural networks.
- High volume of data collected via the internet: « Big data »



What is data?

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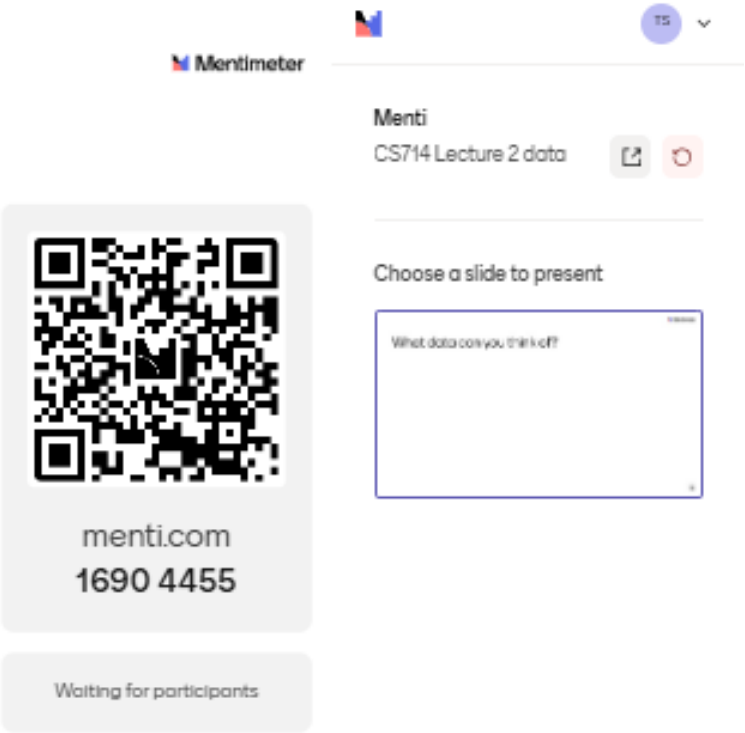
What data can you think of?



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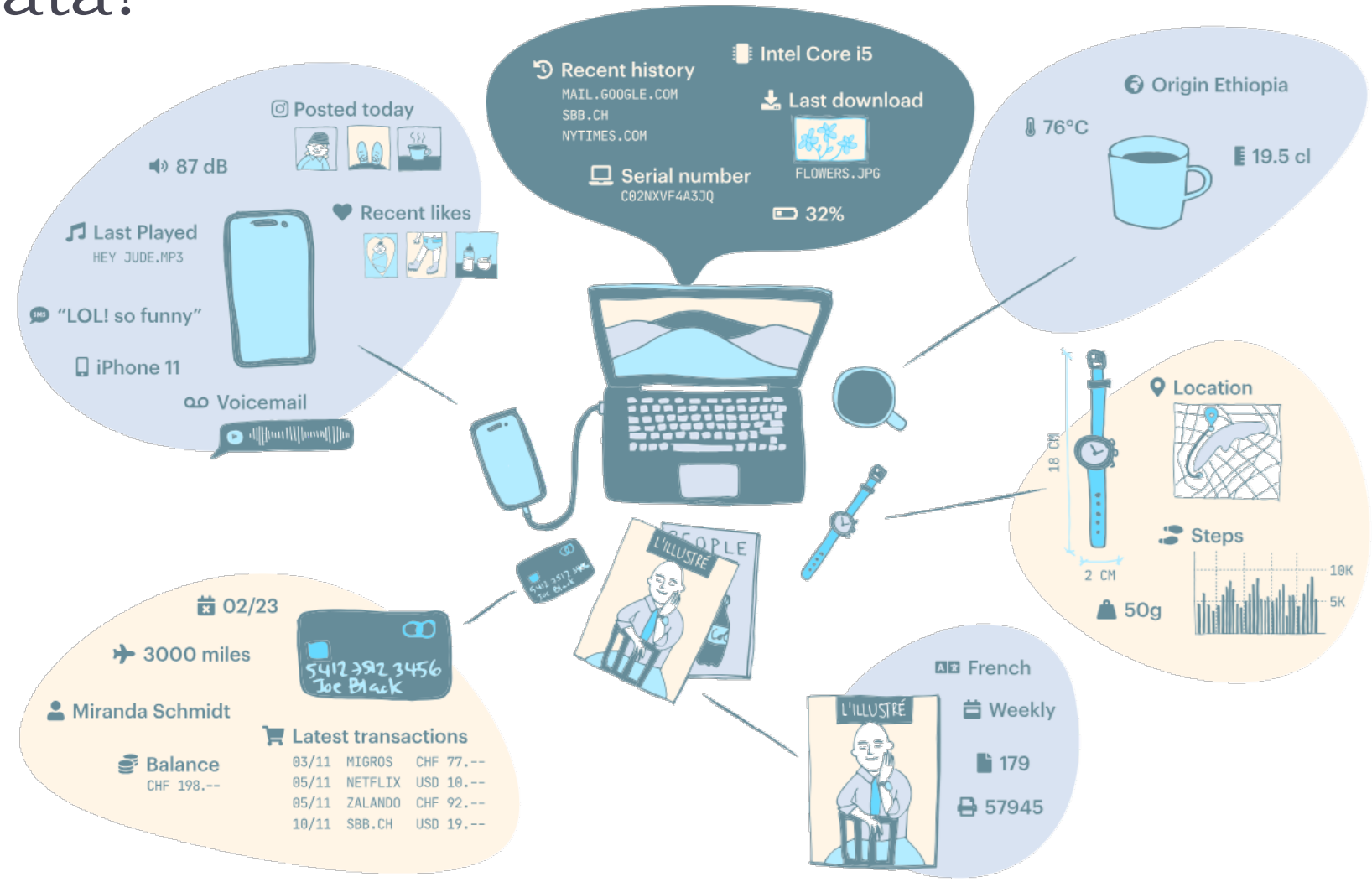

Image Source: https://www.thats-ai.org/en-GB/units/data-comes-in-many-forms

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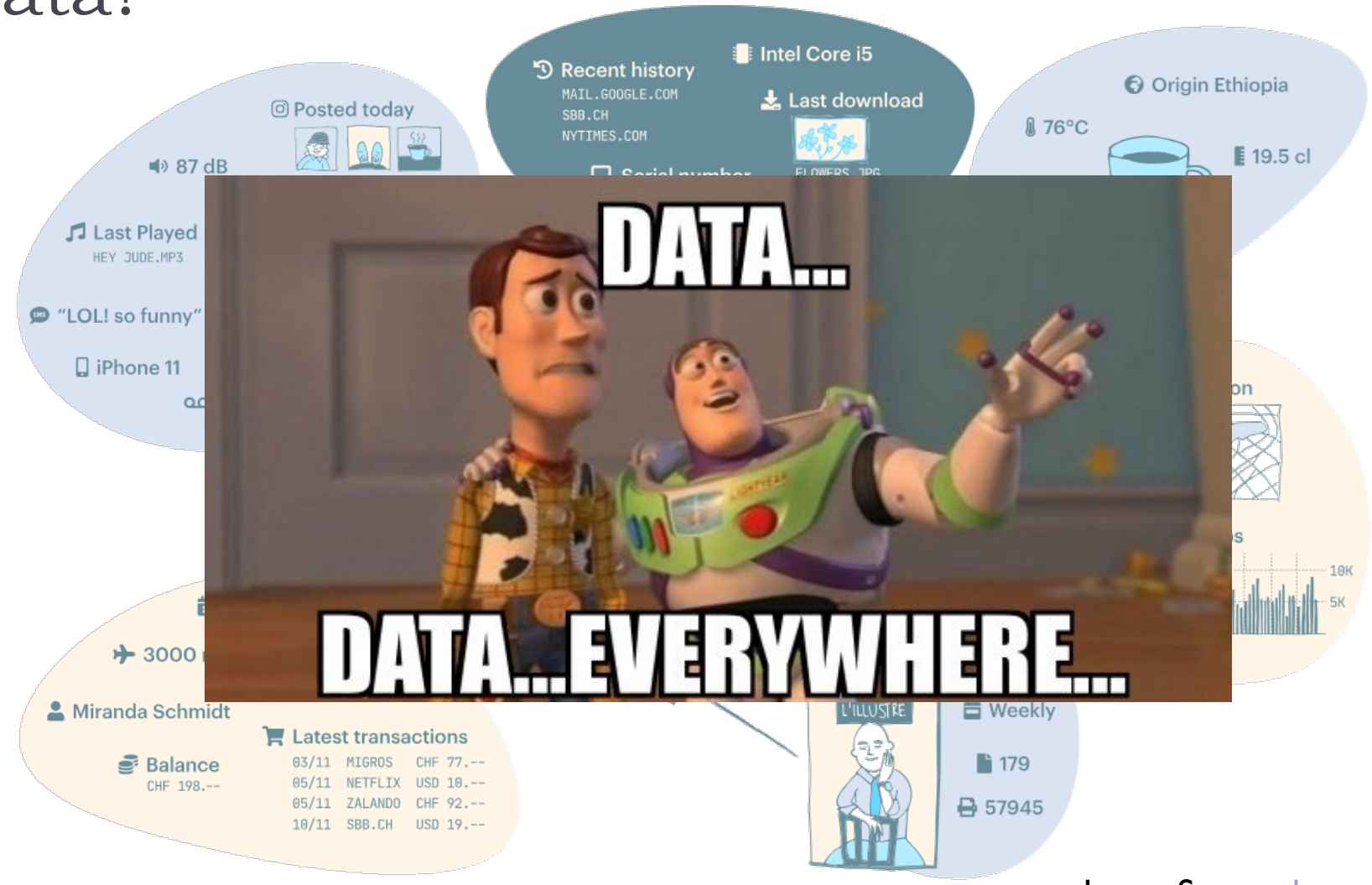


Image Source: https://www.thats-ai.org/en-GB/units/data-comes-in-many-forms

Selecting data

- Selecting relevant data to train a model depends on the task at hand, but also on factors such as:
  - data availability,
  - data quality,
  - ethical and legal constraints,
  - practical considerations like cost, time, and computational resources.

- Example:
  - Consider training a model to detect pedestrians for a self-driving car.
  - What data to choose?
    - Images, videos, radar, lidar?
    - Other factors to consider:
      - Environment (day/night/rain, urban/rural, with pedestrians?)
      - Legal and ethical constrains on data collection and data use (e.g., privacy laws on street recording? copyright?)
      - Cost and computational constrains

Structured vs Unstructured vs Semi-structured data

Data category	Structured data	Unstructured data	Semi-structured data
Definition			
Key characteristics			
Examples			

Structured vs Unstructured vs Semi-structured data

Data category	Structured data	Unstructured data	Semi-structured data
Definition	• Data organised according to a predefined schema, usually in rows and columns		
Key characteristics	• Easy to store, query, and analyse; • Consistent format; • Typically numerical or categorical data.		
Examples	• Spreadsheets, SQL databases, customer records, transaction logs, sensor readings in tables		

Structured vs Unstructured vs Semi-structured data

Data category	Structured data	Unstructured data	Semi-structured data
Definition	• Data organised according to a predefined schema, usually in rows and columns	• Data not following a fixed or predefined structure	
Key characteristics	• Easy to store, query, and analyse; • Consistent format; • Typically numerical or categorical data.	• More flexible but harder to analyse and use directly • Often high-dimensional • Requires more preprocessing	
Examples	• Spreadsheets, SQL databases, customer records, transaction logs, sensor readings in tables	• Text documents, emails, images, videos, audio recordings, social media posts	

Structured vs Unstructured vs Semi-structured data

Data category	Structured data	Unstructured data	Semi-structured data
Definition	• Data organised according to a predefined schema, usually in rows and columns	• Data not following a fixed or predefined structure	• Data not following a rigid tabular structure but contains explicit organisation through tags or key-value pairs
Key characteristics	• Easy to store, query, and analyse; • Consistent format; • Typically numerical or categorical data.	• More flexible but harder to analyse and use directly • Often high-dimensional • Requires more preprocessing	• Flexible schema, • Partially organized, • Easier to parse than unstructured data.
Examples	• Spreadsheets, SQL databases, customer records, transaction logs, sensor readings in tables	• Text documents, emails, images, videos, audio recordings, social media posts	• JSON or XML files, HTML pages, log files, NoSQL databases

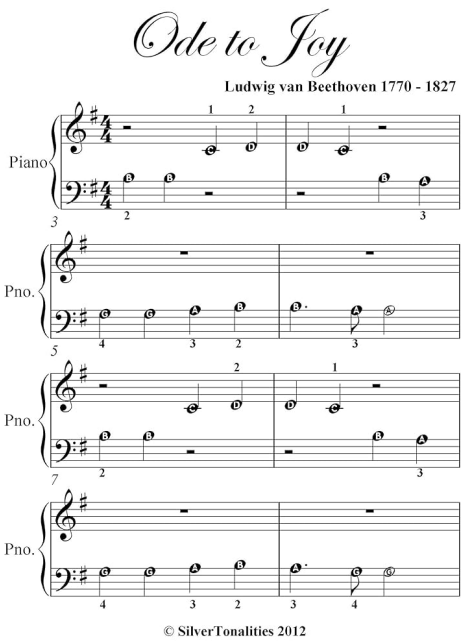
Example of structured tabular data

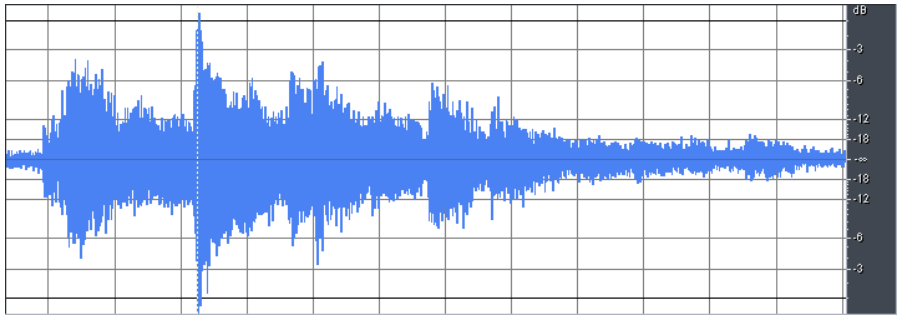
Index	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600.0	NEAR BAY
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500.0	NEAR BAY
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100.0	NEAR BAY
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300.0	NEAR BAY
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200.0	NEAR BAY

Types of attributes

- 2 main types of data attributes in structured data:
  - Numerical: typically used to store quantitative data
    - Continuous (can take any numeric value, e.g., age)
    - Discrete (can take only a finite number of specific values, e.g., number of own cars)
  - Categorical: typically used to store quantitative or qualitative data
    - Nominal (data can fit different categories, e.g., animals being mammal, fish, reptile, etc.)
    - Ordinal (data fits in ranked/ordered categories, e.g., bad/average/good/excellent)
    - Binary (two options, e.g., True/False)

Examples of unstructured data





Example of semi-structured data: JSON file

```
{
  "username": "tech_lover92",
  "timestamp": "2025-12-16T14:32:00Z",
  "text": "Excited about the new AI features in the latest phone update!",
  "hashtags": ["#AI", "#TechNews"],
  "media": [
    {
      "type": "image",
      "url": "https://example.com/image1.jpg"
    }
  ],
  "location": "Auckland, NZ",
  "reactions": {
    "likes": 120,
    "shares": 15,
    "comments": [
      {"user": "ai_fan", "comment": "Can't wait to try it!"},
      {"user": "tech_guru", "comment": "Looks amazing!"}
    ]
  }
}
```

What data does this file represent?

Where can you get data?

- Repositories (e.g., Kaggle, GitHub, OpenML, HugginFace, etc.)
- Python Packages (Scikit-Learn datasets, PyTorch/TensorFlow datasets, etc.)
- Your own collected data (you will need to shape it yourself)
- Web scrapping
- Commercial datasets

→**Important: always check if you can use the data (rights, confidential attributes, etc.) and if you need to take any precautions (e.g., anonymisation).**

→**I will share a Notebook about personal data you can have a look at as extra material.**

Loading data and basic exploration

Dataset characteristics

- After you load a dataset, start by looking at its global characteristics:
  - What is the type of data? (e.g., tabular, images, text, …) / How is it organised?
  - What is the type of the target? (i.e., is it a classification or regression task?)
  - *Classification*: How many classes are they? / *Regression*: What is the target's range of values?
  - How many instances are there in total?
  - *Classification*: How many instances are there per class? (i.e., is there some class imbalance?)
  - What is the type of each attribute?
  - How many attributes are there?

Are you sure your data contains data?

- Remember that machine learning algorithms are trying to build models based on statistical patterns in the data, and relationship between inputs and outputs.

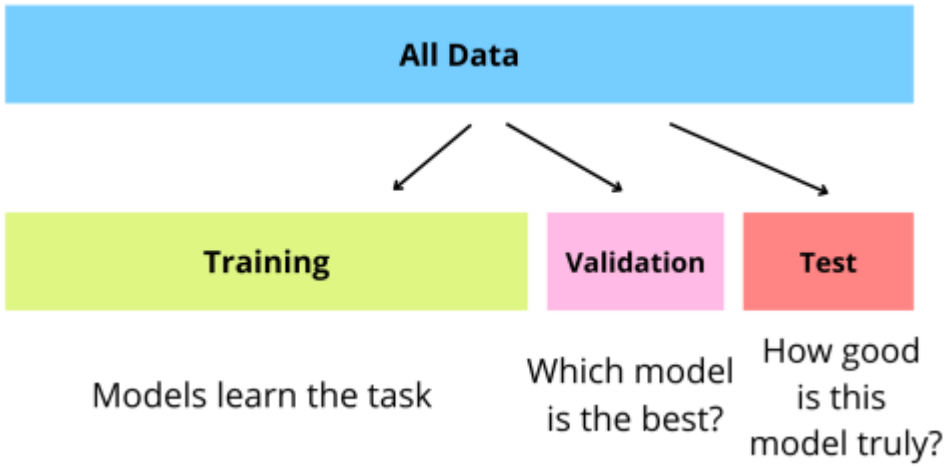
- There might be situations where your data does not contain much “signal” to be used to build a model to answer a question. Therefore, before starting to train a model, you should answer the question "Does my data contain the potential to answer this question?"

- For example:
  - Predicting Completely Random Events (e.g., model the outcome of a coin toss)
  - Inscrutable tasks (e.g., some human behaviour can be very hard to model, even with tons of data)
  - Irrelevant features (e.g., predicting NFT price based on image features)

Before exploring the data further

- To answer the previous question, you need to explore the data further.

- But before this, you need to split your data.



- Very important to do the exploration only on the training set, to avoid data leakage.

Data imbalance

- Data imbalance happens when one or several classes/category are overrepresented (majority classes) in the data, while some other classes are underrepresented (minority classes).
- Examples:
  - *Unbalance in the target classes*: you want to classify images into 2 classes, cat and dog. Your dataset has 100 000 images in total, 95000 images of cats and 5000 image dogs.
  - *Unbalance in a predictive attribute*: you want to predict the probability of somebody getting a loan based on their age, gender, salary, etc. and you have 95% of your data with gender = male.

Degree of imbalance	Proportion of Minority Class
Mild	20-40% of the data set
Moderate	1-20% of the data set
Extreme	<1% of the data set

Table source: Google ML course

Why and when is it a problem?

- Many classification and regression problems are imbalanced! It is hard, sometimes impossible, to collect balanced data (e.g., fraud detection).

- When it can be a problem:
  - When splitting your data in different sets (e.g., train, validation, test sets):
    You need to keep the same distributions for the target and important attributes in your different sets.
  - When training a model:
    Your model might become very good at predicting the majority class(es), but not the minority class(es).
  - When getting insights from a trained model:
    Because of the previous point, the insight you get from your model once trained might not reflect the actual patterns.

How to alleviate imbalance

- There exists several strategies to alleviate imbalance depending on the situation:
  - When splitting your data in different sets (e.g., train, development, test sets):

    Use special sampling approaches such as stratified sampling to sample the sets while maintaining the data distribution.
  - When training a model:

    Use appropriate “imbalance aware” metrics such as precision and recall to evaluate the performance of your model (we will cover that in a future lecture).
  - Acting directly on the data:

    Use upsampling or downsampling, possibly combined. However, these should be used carefully as you might end up losing important information or introducing unwanted bias. Not recommended in small datasets.

How to alleviate imbalance

- Further optional reading about class imbalance strategies: Class Imbalance Strategies — A Visual Guide with Code, Travis Tang, Medium

Exploring the data further

- After having a look at the overall characteristics of the data, you can push the exploration further:

Type of data	Examples of possible actions
Tabular numerical data	• Display scatter plots, histograms, etc., to visualise your data (can use some dimensionality reduction methods if your data has a lot of attributes/dimensions) • Display correlation matrices and plots to visualise relationship between data attributes • Try attribute combinations
Tabular categorical data	• Display the different categories to identify if they are nominal or ordinal
Images	• Display some images for each class • Calculate image sizes and aspect ratios
Text	• Display some text samples • Calculate some text statistics and visualise them with histograms • Depending on the task, you could perform a sentiment analysis, display a word cloud, etc.

- A good understanding of your data is useful to make cleaning and pre-processing choices and to guide the design of your model and training process.

Splitting your data in train and test sets & Further exploration

Data cleaning

- One of the most time-consuming task when preparing data is to clean it.

- Cleaning steps may involve:
  - Removing duplicates (e.g., duplicate images)
  - Fixing formatting issues (e.g., inconsistent floating-point formatting)
  - Dealing with missing values
  - Dealing with outliers

- You can find a more detailed checklist on datacamp.

- Widely used datasets are usually already cleaned when you load them.

Missing values

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Missing values

Note: Analysing why data is missing can be useful as some missing patterns carry important information.

Possible strategies involve:

- Removing data
  - **Removing an instance** (row): e.g. several attribute values are missing
  - **Removing a full attribute** (column): e.g., many entries are missing

- Imputation (filling missing values)
  - **Mean/Median Imputation** (numerical data)
  - **Most Frequent Value (Mode) Imputation** (categorical or numerical data)
  - **Forward/Backward Fill**: Use previous or next values to fill missing entries (common in time series)
  - **Predictive Imputation**: Use a model (e.g., regression or k-NN) to estimate missing values based on other features.
  - **Domain-specific values**: Fill in missing data with values that make sense in the specific context (e.g., zero, "Unknown").

Missing values

What would you do in this example?

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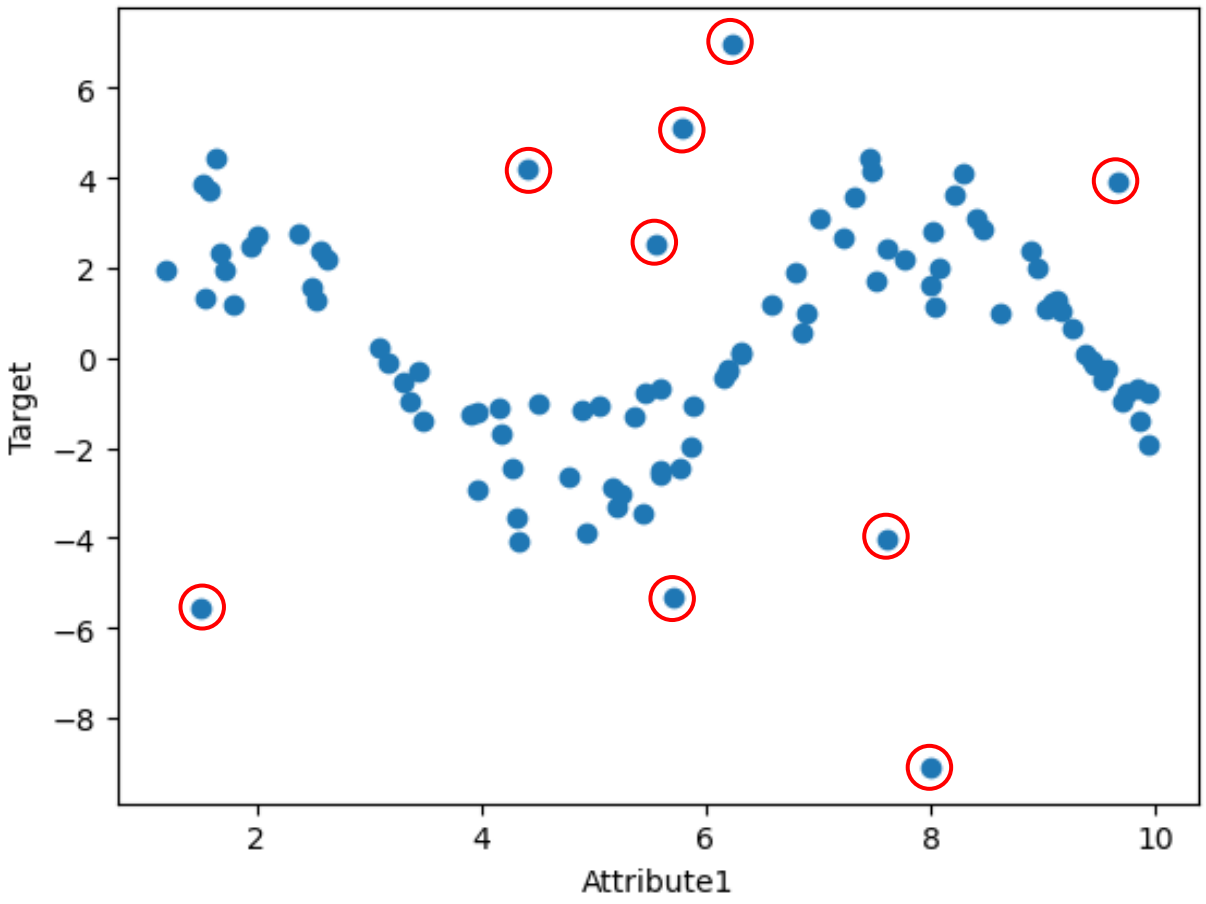
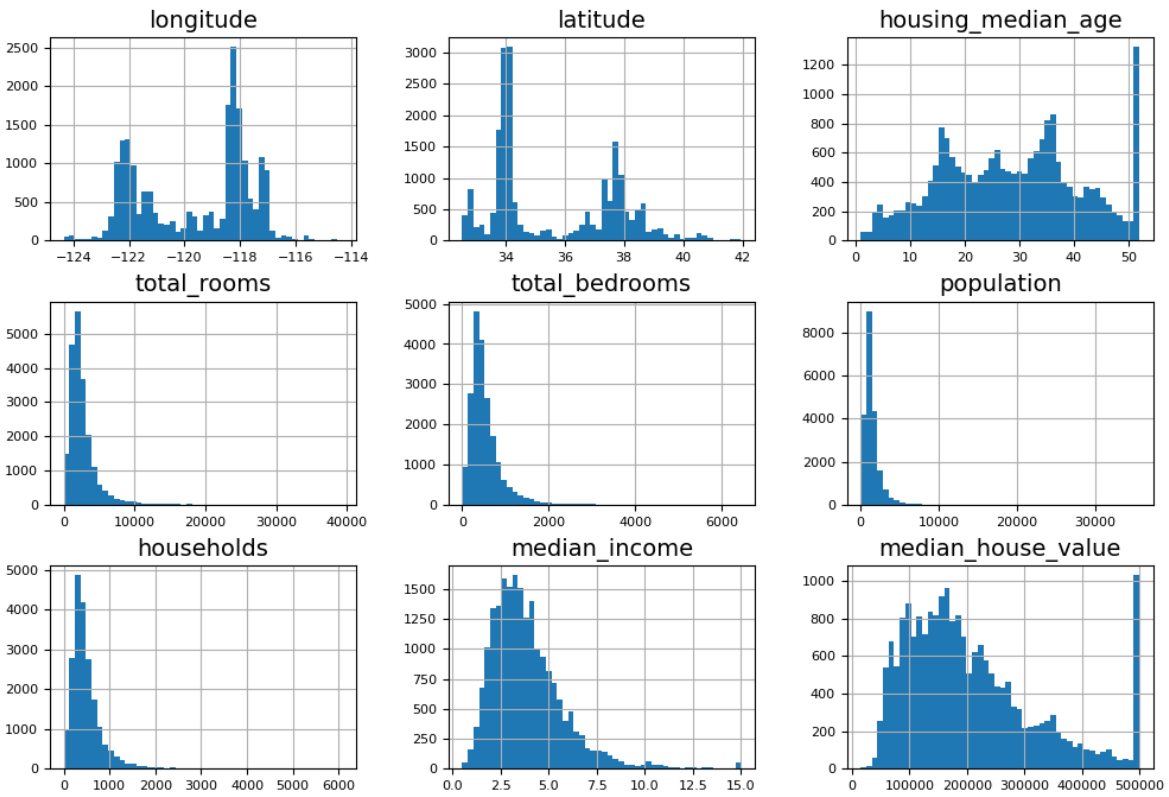
Outliers

- Outliers are data points which are « out of pattern ».
- Can be due to measurement error, data corruption, or can be true outliers.
- Not part of the pattern your model should describe.
- There are a lot of ways to detect outliers, including statistical and ML approaches (e.g., Z-score, Isolation Forest, One-class SVM, etc.)



Further data pre-processing



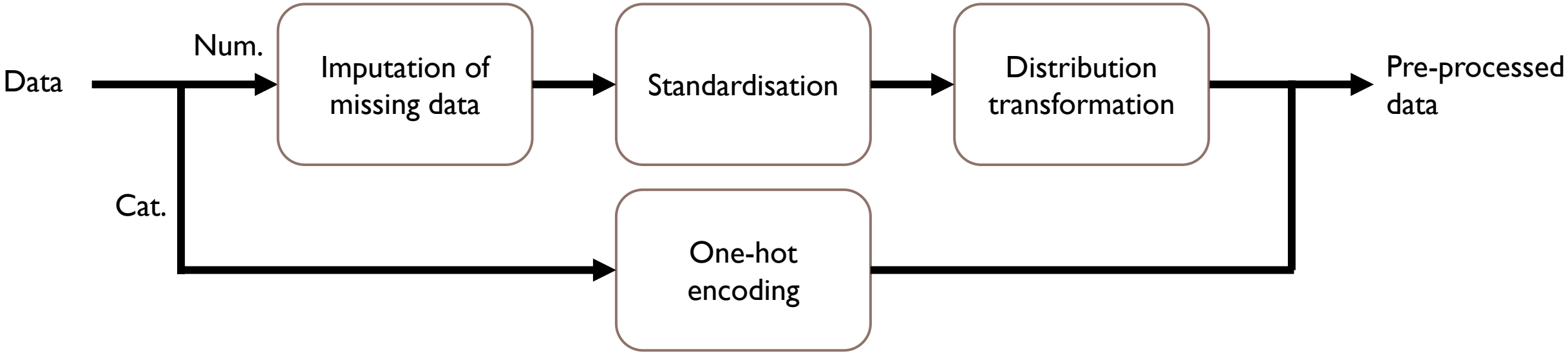
- Making the data suitable for training.

- Further data pre-processing may include:
  - Converting text and categorical attributes to numerical attributes (e.g., one-hot encoding, embedding)
  - Scaling the data (normalisation or standardisation)
  - Applying selected transformations to the data distribution (if not close to Gaussian)
  - Undersampling / oversampling data (if imbalanced)
  - Reducing the number of dimensions
  - Etc.

Data Type	Preprocessing Step	Purpose
Structured (tabular)	Handle missing values (imputation, removal)	Ensure data completeness and quality
	Handle outliers	Remove extreme values that can distort analysis
	Scaling / normalization	Make features comparable in magnitude
	Encoding categorical variables	Convert categorical data into numerical format suitable for models
Semi-structured	Parse and extract relevant fields	Extract meaningful information from semi-structured formats
	Handle missing or inconsistent entries	Improve data quality and consistency
	Standardize formats	Ensure uniformity across entries
Unstructured – Text	Tokenization	Break text into manageable units (words/tokens)
	Stop-word removal	Remove common words that add little value
	Stemming / Lemmatization	Reduce words to their root form for consistency
	Vectorization (TF-IDF, embeddings)	Convert text into numerical features for machine learning
Unstructured – Image	Resizing / cropping	Standardize image dimensions
	Normalization / scaling	Scale pixel values for model stability
	Denoising	Reduce noise to improve model learning
Unstructured – Audio	Resampling / normalization	Standardize sampling rate and signal amplitude
	Noise reduction	Remove unwanted background noise

Building Transformation Pipelines

- In practice: all the pre-processing steps embedded into a **pipeline**, which can be reused on data as needed.
- Pipeline designed and fitted **using the training data** (important to avoid data leakage) and then can be used to process test set and new instances.



Splitting your data in train and test sets & Further exploration

Conclusion

ONE DOES NOT SIMPLY
START TRAINING A MODEL ON RAW DATA
imgflip.com